Tirai — metrics & validation evidence

What is measurably true today, how to verify each number yourself, and what is honestly not yet proven.

HackCanton Season #2 · Financial Applications · solo builder: Pugar Huda Mantoro · repo **github.com/PugarHuda/tirai** · live desk **tirai.vercel.app**

LIVE ON CANTON DEVNET — THE LEDGER IS THE EVIDENCE

> **41 settled trades + 5 atomic multi-leg baskets** — real contracts on the shared 5N Devnet validator, package `tirai-desk 4b1e408f`…, parties `tirai-v1-*`.
>
> **16 best-execution attestations** — each with both dealers' asks disclosed to the regulator *before* settlement, so the clearing price is provably at or below every competing ask.
>
> **32 quote disclosures · 5 open RFQs, each with two sealed quotes** — a live book, not a single demo trade.
>
> **Coverage:** ~25 instruments across sovereigns (US, UK, Germany, Japan, France, Australia, Netherlands, Switzerland, Korea, Singapore, India, Indonesia), supranationals (IBRD, EIB, KfW) and corporates (Amazon, Nvidia, Goldman, JPM), traded on all three rails — reverse-Vickrey (including 3-dealer auctions), direct OTC, and partial fills.

THE PRIVACY CLAIM, VERIFIED ON THE NETWORK (NOT ASSERTED)

The central claim of this project is falsifiable, and we falsify it on every run. `node scripts/devnet.mjs verify` queries the live active-contract set of each party and asserts:

- each dealer sees **only its own quotes** — never a rival's;
- the regulator sees **zero pre-trade contracts** — no RFQs, no quotes;
- the regulator does see **every settled trade** — 46 settlements in its audit view.

It exits non-zero if a single quote ever leaks, so this is a test, not a dashboard. The same result is reproduced independently through the read-only MCP server (`party_view`) and through the hosted desk's "Verify privacy" view, which counts live contracts in the browser.

ENGINEERING EVIDENCE

Check	Result	How to run it
Daml model test suite	**36 / 36 scripts green**	`cd test; daml test`
Hosted QA across three browsers	**66 / 66 checks** (chromium, firefox, webkit)	`node scripts/e2e-hosted.mjs`

MCP server against live Devnet	**25 / 25 checks**	`node scripts/e2e-mcp.mjs`
Read-only proxy security self-test	**14 / 14**, including a party-enumeration bypass regression	`node scripts/test-readonly-proxy.mjs`
On-ledger privacy assertion	**green** after every seeding run	`node scripts/devnet.mjs verify`
Write protection in production	every `/v2/commands/*` call **403**	hosted proxy, verified in production

HYPOTHESES, AND WHAT THE BUILD ACTUALLY VALIDATED

Hypothesis	Status	Evidence
Canton's sub-transaction privacy alone is enough — no ZK, TEE or FHE needed for a confidential RFQ desk	**Validated**	Zero cryptographic machinery in the codebase; privacy asserted on the live network, where a rival dealer's participant node simply never receives the contract
Confidential pre-trade and provable post-trade can coexist	**Validated**	16 attestations built from quotes disclosed to the regulator on demand, with the quotes never public
One settlement integration can serve every asset (cETH, CBTC, CC, USDCx)	**Validated in test**	The cash instrument is any {admin, id}; a second, differently-administered mock registry settles through the identical code path (`testCbtcDvp`)
The desk is usable by an institution's own automation, not just humans	**Validated**	Six MCP tools, 25/25 green against live Devnet, including a write path that posts a real RFQ on-ledger
Desks will pay a per-trade fee for confidentiality	**Not yet validated**	No paying customer. This is the first thing a design-partner pilot must test

REPRODUCE EVERY NUMBER IN THIS DOCUMENT

- **Live desk:** tirai.vercel.app — the tiles, audit trail and best-execution view read the live ledger; nothing is baked in.
- **Ledger:** package `tirai-desk` `4b1e408f6eda27364a55da076d9251ee117f0641f03aaf20883995f1e507a7e3`, parties `tirai-v1-buyer / dealerA / dealerB / regulator / cashissuer / bondissuer`.
- **Code and history:** github.com/PugarHuda/tirai — CI green, plus `JOURNAL.md`, a dated build log of what was built and what broke.

WHAT IS NOT PROVEN YET (STATED PLAINLY)

- **No live cETH or CBTC transaction.** The settlement path is built against the real Splice v1 interfaces and tested against a mock registry, but the Devnet test-token grant has not landed, so no value has moved in the real asset yet.

- **No paying users and no design partner signed.** Every commercial claim in the GTM document is a hypothesis, not a result.
- **Not deployed on the HackCanton participant.** The deploy is written and proven on another Devnet validator; on `hackcanton-01` a wallet-issued token authenticates for reads but DAR upload and party allocation return 403 — that needs participant-admin rights. (Separately: the TLS certificate on the node's Keycloak is expired, which blocks the documented auth path for every participant.)
- **No load or latency benchmarks.** Volumes so far are demonstrative, not performance evidence.

Tirai — Indonesian for "curtain". Price discovery happens behind it. You whisper quotes; the market hears nothing.